



CURSO: MATEMÁTICAS II COD. CURSO: MA-123 MNC

TIPO DE PRUEBA: PRACTICA No. 2 Ex. PARCIAL EX. FINAL EX. SUST.

1. Integrar:

$$\int \frac{6 \csc x}{\sqrt[3]{\sin^3 x + \cos^3 x}} dx \quad (3p)$$

2. Integrar:

$$\int \frac{\sqrt[4]{1+e^{4x}}}{e^x} dx \quad (2p)$$

3. Sea $I_n = \int \frac{1}{(x^3+1)^n} dx$, $\forall n \in \mathbb{Z}^+$

Determinar el valor de A , de tal manera que:

$$I_n = \frac{x}{3(n-1)(x^3+1)^{n-1}} + AI_{n-1}, \quad \forall n \geq 2 \quad (3p)$$

4. Integrar:

$$\int \frac{(1-\sqrt{x^2+x+1})^5}{x^5 \sqrt{x^2+x+1}} dx \quad (3p)$$

5. Integrar:

$$\int \frac{dx}{x^3 \sqrt{x^2+4x-2}} \quad (3p)$$

6. integrar:

$$\int \sqrt[3]{\frac{x^2}{3x+4+\sqrt{48x}}} \cdot \frac{dx}{\sqrt[6]{x^5}} \quad (3p)$$

7. integrar:

$$\int \frac{dx}{x^8 \sqrt{x^3} \sqrt{(x-\sqrt[4]{x})^3}} \quad (3p)$$